

Problem 1:

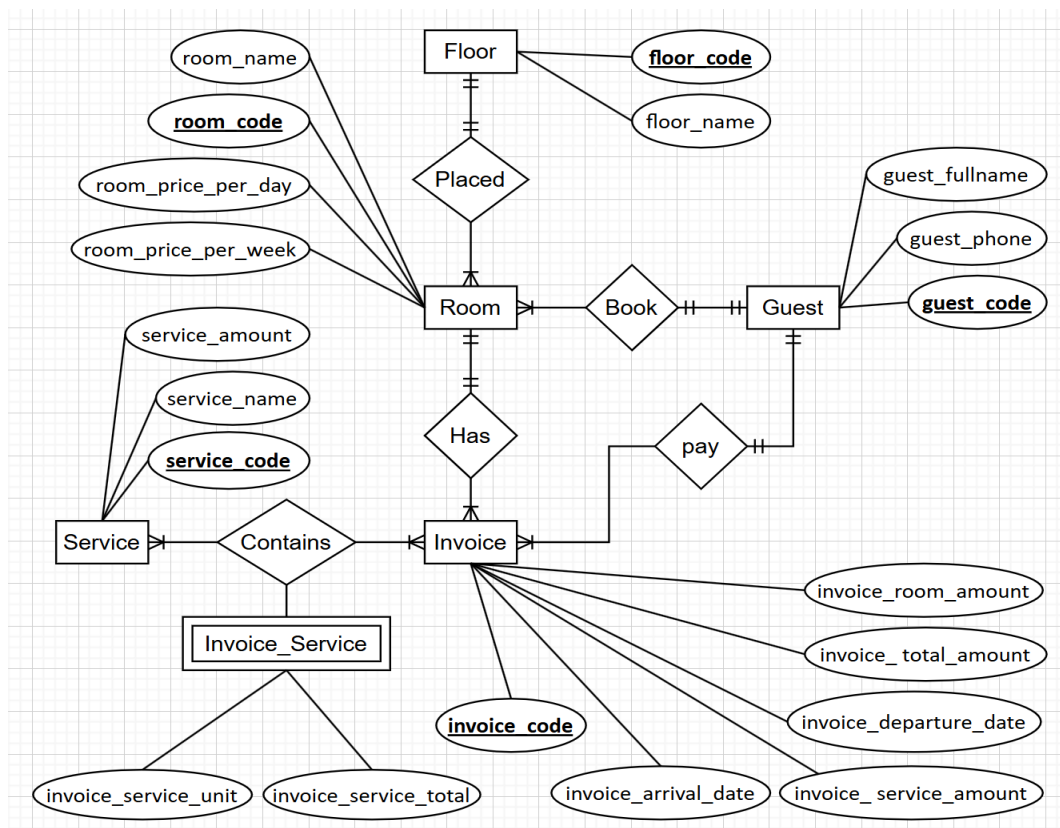
Entity:

- Room: room_code, room_name, room_price_per_day, room_price_per_week, floor_code
- Floor: floor_code, floor_name
- Guest: guest_code, guest_fullname, guest_phone
- Service: service_code, service_name, service_amount
- Invoice: invoice_code, invoice_arrival_date, invoice_departure_date, invoice_room_amount, invoice_service_amount, invoice_total_amount, guest_code, room_code.
- Invoice_Service: invoice_service_unit, invoice_service_total, invoice_code, service_code

Relationship:

- Room – Floor: (N, 1).
- Room – Guest: (1, N)
- Invoice – Service: (M, N)
- Room – Invoice: (N, 1).
- Guest – Invoice: (N, 1)

ER:



Canonical Cover:

- Room:
 - room_code → room_name

- room_code → room_price_per_day
- room_code → room_price_per_week
- room_code → floor_code
- Floor:
 - floor_code → floor_name
- Guest:
 - guest_code → guest_fullname
 - guest_code → guest_phone
- Service:
 - service_code → service_name
 - service_code → service_amount
- Invoice:
 - invoice_code → invoice_arrival_date
 - invoice_code → invoice_departure_date
 - invoice_code → invoice_room_amount
 - invoice_code → invoice_service_amount
 - invoice_code → invoice_total_amount
 - invoice_code → guest_code
 - invoice_code → room_code
- Invoice_Service:
 - (invoice_code, service_code) → invoice_service_unit
 - (invoice_code, service_code) → invoice_service_total

Problem 2:

- a) Find employees who work in room 4.

$$\Pi_{employeeID,fullname}(\sigma_{departmentID=4}(employees \bowtie department))$$

another way: $\sigma_{departmentID=4}(employees)$

- b) Find employees with salaries above 30000.

$$\Pi_{employeeID,fullname}(\sigma_{salary>30000}(employees))$$

- c) For each department, indicate the department name and room location.

$$\Pi_{departmentName,address}(department \bowtie departmentAddress)$$

- d) The average salary of all female employees.

$$g_{avg}(salary) \text{ as } average_salary(\sigma_{gender='Nu'}(employee))$$

- e) Find the names and addresses of all employees of the "Nghien Cuu" department.

$$\Pi_{fullname,address}(\sigma_{departmentName='Nghien Cuu'}(employees \bowtie department))$$